05. 【解答】

1）当文件中的 n 个记录升序排列时，只生成一个归并段，长度达到最大，为 n。若初始工作区内的 m 个元素都大于输入文件中剩下的所有记录，则第一个归并段的长度就为 m，此时为第一个归并段长度最小的情况。排序过程如下表所示。

输出文件 FO	工作区 WA	输入文件 FI
—	—	51, 94, 37, 92, 14, 63, 15, 99, 48, 56, 23, 60, 31, 17, 43, 8, 90, 166, 100
—	51, 94, 37, 92	14, 63, 15, 99, 48, 56, 23, 60, 31, 17, 43, 8, 90, 166, 100
37	51, 94, 14, 92	63, 15, 99, 48, 56, 23, 60, 31, 17, 43, 8, 90, 166, 100
37, 51	94, 14, 92, 63	15, 99, 48, 56, 23, 60, 31, 17, 43, 8, 90, 166, 100
37, 51, 63	15, 94, 14, 92	99, 48, 56, 23, 60, 31, 17, 43, 8, 90, 166, 100
37, 51, 63, 92	15, 94, 14, 99	48, 56, 23, 60, 31, 17, 43, 8, 90, 166, 100
37, 51, 63, 92, 94	15, 14, 99, 48	56, 23, 60, 31, 17, 43, 8, 90, 166, 100
37, 51, 63, 92, 94, 99	15, 14, 56, 48	23, 60, 31, 17, 43, 8, 90, 166, 100
37, 51, 63, 92, 94, 99#	15, 14, 56, 48	23, 60, 31, 17, 43, 8, 90, 166, 100
14	15, 23, 56, 48	60, 31, 17, 43, 8, 90, 166, 100
14, 15	60, 23, 56, 48	31, 17, 43, 8, 90, 166, 100
14, 15, 23	60, 31, 56, 48	17, 43, 8, 90, 166, 100
14, 15, 23, 31	60, 17, 56, 48	43, 8, 90, 166, 100
14, 15, 23, 31, 48	60, 17, 56, 43	8, 90, 166, 100
14, 15, 23, 31, 48, 56	60, 17, 8, 43	90, 166, 100
14, 15, 23, 31, 48, 56, 60	90, 17, 8, 43	166, 100
14, 15, 23, 31, 48, 56, 60, 90	166, 17, 8, 43	100
14, 15, 23, 31, 48, 56, 60, 90, 166	100, 17, 8, 43	—
14, 15, 23, 31, 48, 56, 60, 90, 166#	100, 17, 8, 43	—
8	100, 17, 43	—
8, 17	100, 43	—
8, 17, 43	100	—
8, 17, 43, 100	—	—
8, 17, 43, 100#	—	—

生成三个初始归并段，分别是 37, 51, 63, 92, 94, 99；14, 15, 23, 31, 48, 56, 60, 90, 166；8, 17, 43, 100。

2）最大值为 n，最小值为 m。